Medication-Safe Dual-Encoder Retrieval for Structured Dental Report Prediction from CBCT

Lai Wang[1*], Changpeng Yue[1], Wentao Bao[1], and Junhao Shao[1]

Hangzhou Dianzi University, Hangzhou, China
melo@hdu.edu.cn; ycp93895413@gmail.com; leokia2233@gmail.com; 242270076@hdu.edu.cn

Abstract. MMDental Task 3 predicts seven clinical-report fields from 3D dental CBCT with only 50 labeled reports. We formulate this small-data problem as constrained field retrieval rather than free-text generation, so each output field is traceable to one labeled training report. The proposed framework combines a MedicalNet 3D ResNet-34 with a SwinUNETR-derived encoder. Image-only contrastive adaptation uses all 250 official training CBCTs, including the unlabeled subset, before supervised learning maps each encoder representation to seven field-specific slices of a 168-D latent report vector. At inference, each branch independently ranks the fixed field-specific candidate banks using visual and predicted-report similarity. A medication-preserving gate fixes Diagnosis and Handle to the MedicalNet branch and admits a Swin candidate for another field only when the two candidates contain the same recognized medication-pattern set. Official online evaluation yields a `Weighted_Score` of 0.235259, diagnosis-code F1 of 0.161000, treatment-action F1 of 0.346893, medication F1 of 0.470000, and complete nonempty output for all evaluated cases. The results support auditable retrieval as a practical alternative to unconstrained generation in a highly limited supervision regime. The gate preserves recognized lexical mentions but does not certify clinical safety.

Keywords: Dental CBCT · Structured Report Prediction · Cross-modal Retrieval · Medication Preservation · Self-supervised Learning

1 Introduction

CBCT provides 3D dentoalveolar information for dental diagnosis and treatment planning, yet converting a scan into a structured clinical record requires linking anatomy to findings, procedures, tooth notations, and medications [13]. MMDental Task 3 defines seven fields: Main appeal, Present medical history, Oral Check, Diagnosis, Treatment plan, Handle, and Doctor advices. It releases 50 labeled CBCT–report pairs, 200 unlabeled CBCTs, and 50 public Validation scans with hidden reports [16]. Codabench combines lexical-overlap, tooth, diagnosis, action, medication, and completion metrics into a `Weighted_Score` [19].

* Corresponding author.

The outputs studied here are challenge predictions, not substitutes for clinician judgment.

Medical report generators can produce coherent text from image features and learned memories [6,3], but 50 reports provide little evidence for rare codes and medications. Retrieval restricts the output to observed clinical text and preserves field-level provenance, which motivates it over unconstrained generation in this limited-data setting. For volumetric representation, MedicalNet provides a transferable convolutional prior [20,5]. Swin Transformers provide hierarchical context [9]; we use a publicly self-supervised SwinUNETR implementation [14]. The branches' different inductive biases are potentially complementary, but unconstrained fusion can change medication mentions. We therefore use MedicalNet as a conservative anchor and admit the Swin alternative only through a field-level medication-pattern preservation rule. Throughout this paper, "medication-safe" is a compact system name for this lexical constraint, not a claim of clinical safety.

Our constrained dual-encoder pipeline (Fig. 1) learns MedicalNet and Swin representations, retrieves one complete candidate per field from labeled-data candidate banks, and applies a medication-pattern gate. Our contributions are threefold: image-only adaptation is separated from supervised seven-field report learning; field-conditioned cross-modal retrieval copies each field from a traceable labeled case; and an inference-time dual-encoder gate admits complementary Swin retrieval only when it preserves the recognized medication patterns.

2 Method

Our method treats structured report prediction as field-wise retrieval with constrained branch selection, as summarized in Fig. 1. A deterministic transform first maps each CBCT to a three-window volume. MedicalNet and SwinUNETR-derived branches then produce independent visual and latent-report representations. The labeled cases define seven field-specific candidate banks, from which each branch retrieves one complete text per field. Finally, an inference-only medication-pattern gate selects one branch per field and validates the seven-field output. The following subsections follow this data flow; the two training objectives are presented last to separate optimization from inference.

2.1 Problem Formulation and Three-Window Preprocessing

Let $\mathcal{L} = \{(x_i, \mathbf{c}_i)\}_{i=1}^{N_l}$ be the labeled set, where x_i is a CBCT volume and $\mathbf{c}_i = \{c_i^f\}_{f\in\mathcal{F}}$ is its structured report over the seven official fields $\mathcal{F}$. Let $\mathcal{U} = \{u_j\}_{j=1}^{N_u}$ be the unlabeled CBCT set. Given a query x_q, the system must return $\hat{\mathbf{c}}_q = \{\hat{c}_q^f\}_{f\in\mathcal{F}}$. Every field is copied from one nonempty labeled candidate before deterministic shortening and schema validation; fields may select different source cases, but text from multiple candidates is never combined within one field.

All volumes are converted to a common anatomical axis order, resampled to a fixed spatial resolution, and mapped to three intensity windows. We denote the resulting tensor by $V(x)$. The transform is deterministic for supervised

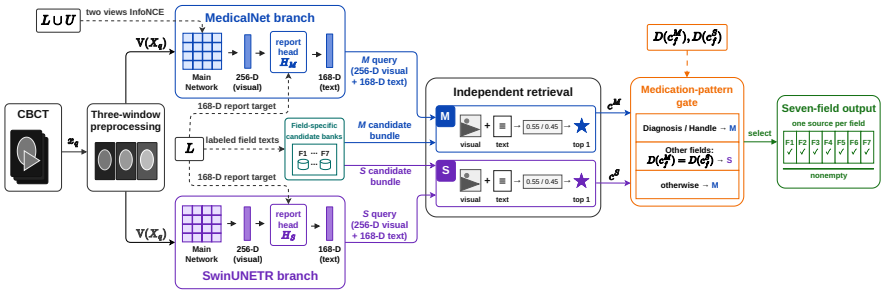


Fig. 1. Medication-preserving dual-encoder retrieval. Dashed connectors denote image adaptation, report supervision, or labeled-data-derived control; solid connectors denote inference. The independent retrieval module receives four bundled inputs: MedicalNet and Swin query bundles, each containing a 256-D visual vector and a predicted 168-D report vector, and two branch-specific candidate bundles from the fixed field banks. Within each field, the two lanes independently combine visual and report-space similarities and emit top-1 candidates. For compactness, the lane outputs labeled c^M and c^S in the artwork denote the field-indexed candidates c_f^M and c_f^S. The gate fixes Diagnosis and Handle to MedicalNet and uses Swin for another field only when $D(c_f^M) = D(c_f^S)$; one complete, nonempty text is emitted per field.

learning and inference; stochastic views are introduced only for MedicalNet's image-only adaptation. Fixed complementary windows expose different intensity ranges without requiring a learned preprocessor from only 50 labeled cases.

2.2 Independent Visual and Latent-Report Branches

The convolutional branch uses a MedicalNet-initialized 3D ResNet-34 [20,5]. Before report supervision, its encoder E_M is adapted using two stochastic views of every image in $\mathcal{L} \cup \mathcal{U}$, without using any report text. This is the semi-supervised component of the pipeline. The complementary branch retains only the encoder of a publicly self-supervised SwinUNETR [9,14]; it has no segmentation decoder and does not participate in the two-view adaptation. After global aggregation and projection, branch $b \in \{M, S\}$ produces the 256-D visual representation

$$\mathbf{z}_i^b = E_b(V(x_i)) \in \mathbb{R}^{d_v}, \qquad d_v = 256,$$

where $\mathbb{R}^d$ is the d-dimensional real vector space, and M and S denote MedicalNet and Swin, respectively.

This pairing is deliberate: adaptation on all official CBCTs makes the convolutional branch the conservative anchor, while shifted-window attention supplies a structurally different context representation without an additional decoder. The two representations remain independent so that every retrieval decision retains a well-defined branch provenance.

Each branch owns a report head $H_b : \mathbb{R}^{d_v} \to \mathbb{R}^{d_r}$, where $d_r = 168$. The heads share neither parameters nor visual features and predict

$$\hat{\mathbf{t}}_i^b = H_b(\mathbf{z}_i^b), \qquad \hat{\mathbf{t}}_i^{b,f} = \hat{\mathbf{t}}_i^b[f],$$

where $[f]$ denotes the d_f-dimensional slice assigned to field f. During supervised learning, a labeled CBCT is mapped to a predicted concatenated report vector; during inference, the two outputs $(\mathbf{z}_q^b, \hat{\mathbf{t}}_q^b)$ form the visual-plus-report query bundle shown in Fig. 1. The "text" component is a predicted latent vector, not report text supplied with the query.

2.3 Field-Specific Candidate Banks

Only $\mathcal{L}$ supplies report supervision and retrieval candidates. For each field f, the nonempty labeled texts form

$$\mathcal{C}_f = \{c_i^f \mid (x_i, \mathbf{c}_i) \in \mathcal{L},\ c_i^f \neq \varnothing\}.$$

We fit a separate mapping ϕ_f for each field using TF–IDF weighting followed by truncated SVD [7,4]. It maps each text to the normalized target

$$\mathbf{t}_i^f = \phi_f(c_i^f) \in \mathbb{R}^{d_f}.$$

The seven slices are concatenated into the common report-head target

$$\mathbf{t}_i = [\mathbf{t}_i^f]_{f \in \mathcal{F}} \in \mathbb{R}^{d_r}, \qquad \sum_{f \in \mathcal{F}} d_f = d_r = 168.$$

The same $\mathbf{t}_i$ supervises H_M and H_S; the target representation is shared, but the two report heads and their visual inputs remain independent. Empty labeled fields use a fixed nonzero placeholder slice for report-head supervision but are excluded from the retrieval candidates $\mathcal{C}_f$. Separate rather than pooled banks follow the official seven-field schema and prevent a text observed under one field from being returned under another; retaining the complete source text also preserves field-level provenance.

The bank for field f stores, for every eligible labeled source case, its source index, complete field text, field-space target, and the visual embeddings produced by both branches:

$$\mathcal{K}_f = \{(i, c_i^f, \mathbf{t}_i^f, \mathbf{z}_i^M, \mathbf{z}_i^S) \mid c_i^f \in \mathcal{C}_f\}. \tag{1}$$

Thus, the raw candidate text c_i^f and target $\mathbf{t}_i^f$ are shared, while the visual cache is branch-specific. The two outgoing candidate bundles in Fig. 1 are the corresponding views

$$\mathcal{K}_f^M = \{(\mathbf{z}_i^M, \mathbf{t}_i^f, c_i^f)\}_i, \qquad \mathcal{K}_f^S = \{(\mathbf{z}_i^S, \mathbf{t}_i^f, c_i^f)\}_i.$$

They are not separate text libraries. Candidate-bank construction is an offline operation; the banks remain fixed when an unseen query is processed.

2.4 Independent Field-Wise Retrieval

For each field f, the retrieval block receives exactly four bundled inputs. The two query lines carry

$$Q_q^M = (\mathbf{z}_q^M, \hat{\mathbf{t}}_q^M), \qquad Q_q^S = (\mathbf{z}_q^S, \hat{\mathbf{t}}_q^S),$$

and the two candidate lines carry $\mathcal{K}_f^M$ and $\mathcal{K}_f^S$ defined above. A lane uses only its matching query and candidate bundle: the MedicalNet lane receives $(Q_q^M, \mathcal{K}_f^M)$, and the Swin lane receives $(Q_q^S, \mathcal{K}_f^S)$. There is no cross-lane feature, score, or text fusion.

Within branch $b \in \{M, S\}$, the 168-D query is sliced to $\hat{\mathbf{t}}_q^{b,f}$, and candidate i is scored using

$$s_{\text{vis}}^b(q,i) = \cos(\mathbf{z}_q^b, \mathbf{z}_i^b), \qquad s_{\text{rep}}^{b,f}(q,i) = \cos(\hat{\mathbf{t}}_q^{b,f}, \mathbf{t}_i^f),$$

where $\cos(\cdot,\cdot)$ denotes cosine similarity. Their fixed weighted sum is

$$R_f^b(q,i) = \lambda s_{\text{vis}}^b(q,i) + (1-\lambda) s_{\text{rep}}^{b,f}(q,i), \qquad \lambda = 0.55. \tag{2}$$

The fixed 0.55/0.45 weighting gives the query image a slight majority while retaining the field-specific semantic match predicted by the report head. Each lane independently returns

$$i_{q,f}^{b\star} = \arg\max_{i: c_i^f \in \mathcal{C}_f} R_f^b(q,i), \qquad c_f^b = c_{i_{q,f}^{b\star}}^f.$$

Here, $i_{q,f}^{b\star}$ is the selected labeled-case index and c_f^b its complete field text. The retrieval output is therefore one top-1 text per branch and field, together with source and score provenance. Retrieval is an inference operation rather than a training loss.

2.5 Medication-Pattern Gate and Structured Output

Before evaluation, medication-related expressions observed in the labeled training reports are manually audited and fixed as a case-insensitive pattern recognizer. Let $\mathcal{P}$ be the resulting set of canonical patterns, and define

$$D(c) = \{p \in \mathcal{P} \mid p \text{ is recognized in candidate } c\}.$$

Thus, $D(c)$ records only recognized lexical patterns; it does not assess indication, dose, route, negation, or clinical correctness.

The gate operates only at inference, after both branches have independently retrieved their top-1 texts:

$$\hat{c}_q^f = \begin{cases} c_f^M, & f \in \{\text{Diagnosis}, \text{Handle}\}, \\ c_f^S, & f \notin \{\text{Diagnosis}, \text{Handle}\} \ \wedge \ D(c_f^M) = D(c_f^S), \\ c_f^M, & \text{otherwise.} \end{cases} \tag{3}$$

Here, $\wedge$ denotes logical conjunction. Diagnosis and Handle are anchored to MedicalNet because these task fields respectively carry diagnosis codes and medication-bearing procedures. For the other fields, the equality condition permits complementary Swin retrieval only when the recognized medication vocabulary is unchanged. The equality condition prevents a switch that adds, removes, or renames a recognized medication pattern relative to the MedicalNet candidate. It remains a lexical preservation rule, not a clinical safety claim.

Finally, deterministic field-specific length and sentence-unit limits shorten selected candidates without reordering or combining sources. Empty outputs receive a fixed fallback. The system validates the exact seven-field schema and emits one nonempty string per field. Consequently, training ends with two independent retrievers, whereas inference additionally runs the branch-selection gate and structured-output validator.

2.6 Loss Functions

The image-only adaptation and labeled report learning use distinct objectives. For a batch of B CBCTs, let a_i and b_i be two stochastic augmentations of $V(x_i)$, and let $\mathbf{u}_i^a$ and $\mathbf{u}_i^b$ be the normalized temporary projections of $E_M(a_i)$ and $E_M(b_i)$. With temperature τ, the directed contrastive loss is

$$\ell_{a\to b} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp((\mathbf{u}_i^a)^\top \mathbf{u}_i^b/\tau)}{\sum_{j=1}^{B} \exp((\mathbf{u}_i^a)^\top \mathbf{u}_j^b/\tau)}.$$

The symmetric image-adaptation objective is

$$\mathcal{L}_{\text{img}} = \frac{1}{2}\left(\ell_{a\to b} + \ell_{b\to a}\right). \tag{4}$$

The reverse term $\ell_{b\to a}$ is obtained by interchanging the two views. Here, i indexes the anchor CBCT and j indexes denominator candidates; $\cup$ denotes set union. Matched views of the same CBCT are positives and the other batch members are negatives [2]. This objective uses images from $\mathcal{L}\cup\mathcal{U}$, updates the MedicalNet encoder and its temporary projection head, and uses no report label. The projection head is discarded before supervised report learning.

For a labeled minibatch $\mathcal{B}_l \subset \mathcal{L}$, each branch is then trained independently with the mean field-wise cosine distance

$$\mathcal{L}_{\text{report}}^{b} = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}}\left[1 - \frac{1}{|\mathcal{B}_l|}\sum_{i\in\mathcal{B}_l}\frac{(\hat{\mathbf{t}}_i^{b,f})^\top \mathbf{t}_i^f}{\|\hat{\mathbf{t}}_i^{b,f}\|_2 \|\mathbf{t}_i^f\|_2}\right]. \tag{5}$$

Here, $|\mathcal{F}|$ and $|\mathcal{B}_l|$ denote set cardinalities, $(\cdot)^\top$ denotes transpose, and $\|\cdot\|_2$ is the Euclidean norm. All target slices, including the fixed placeholder for an empty field, are nonzero after normalization. The report head is optimized from the start; its encoder is initially frozen and then jointly updated. No cross-branch consistency, retrieval, or medication-gate term is included in either training objective.

Table 1. Local environment; the organizer target was Ubuntu 20.04/RTX 3090.

Item	Recorded configuration
System / CPU / RAM	Debian 12; i7-13700KF (16 cores, 24 threads); 62.6 GiB
GPU / driver	1× RTX 4090, 24 GB; NVIDIA 535.247.01; driver CUDA 12.2
Python / framework	Python 3.10.20; PyTorch 2.5.1+cu121 (CUDA 12.1); MONAI 1.5.1
Data / container	NumPy 2.2.6; pandas 2.3.3; scikit-learn 1.7.2; nibabel 5.4.2; Docker 29.6.2

Table 2. Formal training protocols. SwinUNETR denotes its encoder only; times exclude checking, inference, and packaging.

Setting	MedicalNet branch	Swin branch
Initialization	MedicalNet ResNet-34; InfoNCE on 250 official CBCTs	Public self-supervised SwinUNETR
Shared protocol	50 labeled cases; input 3×112×160×160; AdamW, LR/weight decay 2×10^{-4}, no scheduler; Eq. 5; FP16 AMP; clip 3.0	
Epochs / frozen / batch	300 / first 30 / 4	60 / first 10 / 1
Parameters	63,514,728	8,303,322
Training time / final loss	43.3 min / 0.37809	113.0 min / 0.49855

3 Experiments

Dataset and evaluation. We used 50 Train-Labeled cases for report supervision, 200 Train-Unlabeled CBCTs only for image adaptation, and 50 Validation-Public scans for prediction [16]. The unlabeled subset contributed no report text or structured target. All labeled reports defined the supervised targets and retrieval libraries used by the final system. Codabench reports the lexical metrics BLEU, METEOR, ROUGE-L, and CIDEr [12,1,8,15]. It also reports diagnosis, tooth, action, plan-recall, medication, completion, and composite `Weighted_Score` metrics [19].

Implementation and training protocol. Tables 1 and 2 record the environment and protocols. Both branches used AdamW, AMP, and gradient clipping at 3.0. Supervised training used no spatial augmentation; two-view augmentation occurred only in MedicalNet's image-only adaptation.

4 Results and Discussion

4.1 Official Evaluation

The proposed system achieved a `Weighted_Score` of 0.235259 and complete nonempty output for all 50 official evaluation cases (Table 3).

The modest lexical scores show that retrieval does not ensure close reference wording. Tooth extraction was recall-oriented (0.452952 recall versus 0.065738 precision), while medication and action F1 reached 0.470000 and 0.346893; aggregate evaluation cannot isolate the contribution of an individual component.

4.2 Qualitative Gate Audit

We audited all $50 \times 7 = 350$ decisions reconstructed from the frozen checkpoints. The gate accepted Swin 231 times, rejected it 19 times because medication sets

Table 3. Official evaluation on 50 cases. Rank, per-case scores, and significance are unavailable.

Metric	Score	Metric	Score
`Weighted_Score`	.235259	BLEU-1	.105871
METEOR	.188825	ROUGE-L	.147121
CIDEr	.038662	Diagnosis-code F1	.161000
Treatment-action F1	.346893	Treatment-plan recall	.579333
Medication F1	.470000	Field completion	1.000000
Tooth precision / recall / F1		.065738 / .452952 / .109301	

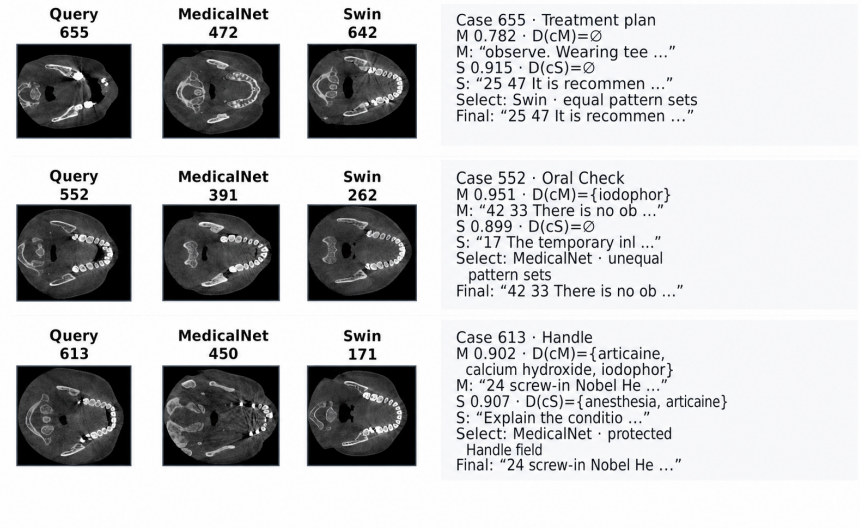


Fig. 2. Medication-gate audit showing query and source CBCTs, candidate excerpts, recognized sets, and the emitted field. Cases 655, 552, and 613 respectively illustrate equal-set Swin selection, unequal-set MedicalNet retention, and the fixed MedicalNet rule for Handle. Images are central axial slices.

differed, and assigned all 100 Diagnosis/Handle decisions to MedicalNet. Figure 2 gives one deterministic example per control path; selection used no hidden report, online case-level score, or manual image interpretation.

Case 552 demonstrates that the gate can prevent removal of a recognized medication pattern, while case 613 confirms that Handle bypasses the equality test. Branch scores are representation-specific similarities, not comparable probabilities. With hidden Validation references, this audit establishes execution and provenance, not clinical correctness.

Efficiency and deployment. On the local RTX 4090, container inference required 77.490 s for 50 cases (1.550 s/case), with a 7,273 MiB sampled peak and approximately 1.003 counted TFLOPs per case; these local measurements do not represent organizer RTX 3090 latency.

Limitations and future work. Fifty reports limit generalization; retrieval may copy mismatched details and cannot ensure cross-field consistency. No directly comparable baseline was evaluated under the identical hidden-reference

protocol, so the reported result demonstrates feasibility rather than superiority or component-wise causality. Medication equality validates neither meaning, indication, dose, route, nor negation; despite the system name, it cannot guarantee clinical safety. Aggregate scores also prevent case-level statistical analysis. Future work should study calibrated abstention, controlled component comparisons, tooth-precision-aware reranking, and entity-constrained editing.

5 Conclusion

Medication-preserving dual-encoder retrieval reached a 0.235259 official composite score with all seven fields completed. Traceable single-source output constrains synthesis, but lexical scores and recall-oriented tooth behavior show that provenance does not ensure semantic or clinical correctness.

Acknowledgements. This work was developed and evaluated within the ODIN 2026 challenge cluster, comprising the STS, ToothFairy, and Bite2Text challenges [18,10,11,17]. We thank the organizers and data owners for making the challenge resources available, and Codabench [19] for hosting the evaluation platform.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005), https://aclanthology.org/W05-0909/ 7
2. Chen, T., Kornblith, S., Norouzi, M., Hinton, G.: A simple framework for contrastive learning of visual representations. In: Proceedings of the 37th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 119, pp. 1597–1607 (2020), https://proceedings.mlr.press/v119/chen20j.html 6
3. Chen, Z., Song, Y., Chang, T.H., Wan, X.: Generating radiology reports via memory-driven transformer. In: Conference on Empirical Methods in Natural Language Processing. pp. 1439–1449 (2020). https://doi.org/10.18653/v1/2020.emnlp-main.112 2
4. Deerwester, S., Dumais, S.T., Furnas, G.W., Landauer, T.K., Harshman, R.: Indexing by latent semantic analysis. Journal of the American Society for Information Science **41**(6), 391–407 (1990). https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<391::AID-ASI1>3.0.CO;2-9 4
5. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: IEEE Conference on Computer Vision and Pattern Recognition. pp. 770–778 (2016). https://doi.org/10.1109/CVPR.2016.90 2, 3
6. Jing, B., Xie, P., Xing, E.: On the automatic generation of medical imaging reports. In: Annual Meeting of the Association for Computational Linguistics. pp. 2577–2586 (2018). https://doi.org/10.18653/v1/P18-1240 2

7. Jones, K.S.: A statistical interpretation of term specificity and its application in retrieval. Journal of Documentation **28**(1) (1972). https://doi.org/10.1108/eb026526 4
8. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Text Summarization Branches Out. pp. 74–81 (2004), https://aclanthology.org/W04-1013/ 7
9. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., et al.: Swin transformer: Hierarchical vision transformer using shifted windows. In: IEEE/CVF International Conference on Computer Vision (2021). https://doi.org/10.1109/ICCV48922.2021.00986 2, 3
10. Lumetti, L., Di Bartolomeo, M., Pellacani, A., Anesi, A., Grana, C., Bolelli, F.: Ontology-grounded structured prediction for dental CBCT reporting. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 (2026) 9
11. Lumetti, L., Rizzo, F., Cremonini, F., Candeloro, E., Luca, L., Grana, C., Bolelli, F.: Do multimodal LLMs understand intraoral dental data? dataset, platform, and baselines. In: European Conference on Computer Vision (ECCV) (2026) 9
12. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Annual Meeting of the Association for Computational Linguistics. pp. 311–318 (2002). https://doi.org/10.3115/1073083.1073135 7
13. Schwendicke, F., Samek, W., Krois, J.: Artificial intelligence in dentistry: Chances and challenges. Journal of Dental Research (2020). https://doi.org/10.1177/0022034520915714 1
14. Tang, Y., Yang, D., Li, W., Roth, H.R., Landman, B.A., Xu, D., et al.: Self-supervised pre-training of swin transformers for 3d medical image analysis. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (2022). https://doi.org/10.1109/CVPR52688.2022.02007 2, 3
15. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: IEEE Conference on Computer Vision and Pattern Recognition (2015). https://doi.org/10.1109/CVPR.2015.7299087 7
16. Wang, C., Zhang, Y., Wu, C., et al.: MMDental - a multimodal dataset of tooth CBCT images with expert medical records. Scientific Data (2025). https://doi.org/10.1038/s41597-025-05398-7 1, 7
17. Wang, Y., Li, Z., Wu, C., Liu, J., Zhang, Y., Ni, J., Luo, Q., Chen, J., Zhang, H., Liu, J., et al.: MICCAI STS 2024 challenge: Semi-supervised instance-level tooth segmentation in panoramic x-ray and CBCT images. Medical Image Analysis p. 103986 (2026). https://doi.org/10.1016/j.media.2026.103986 9
18. Wang, Y., Zhang, Y., Chen, X.D., Wang, S., Qian, D., Ye, F., Xu, F., Zhang, H., Dan, R., Zhang, Q., et al.: MICCAI 2023 STS challenge: A retrospective study of semi-supervised approaches for teeth segmentation. Pattern Recognition **170**, 112049 (2026). https://doi.org/10.1016/j.patcog.2025.112049 9
19. Xu, Z., Escalera, S., Pavão, A., Richard, M., Tu, W.W., Yao, Q., Zhao, H., Guyon, I.: Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform. Patterns **3**(7), 100543 (2022). https://doi.org/10.1016/j.patter.2022.100543 1, 7, 9
20. Zhou, Z., Sodha, V., Pang, J., Gotway, M.B., Liang, J.: Models genesis. Medical Image Analysis **67**, 101840 (2021). https://doi.org/10.1016/j.media.2020.101840 2, 3